

Linear constraints

Degeneracy

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Optimization: principles and algorithms



Degeneracy

*The concepts of vertex and basic feasible solutions are equivalent.
But there is not necessarily a bijection between the two sets.
A vertex may correspond to several basic feasible solutions.*

Example

Consider the polyhedron in $\mathbb{R}^2$:

$$x_1 + x_2 \leq 1$$

$$0 \leq x_1 \leq 1$$

$$0 \leq x_2 \leq 1$$

We consider the equivalent polyhedron in standard form in $\mathbb{R}^5$:

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + x_4 = 1$$

$$x_2 + x_5 = 1$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Example: first basis

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

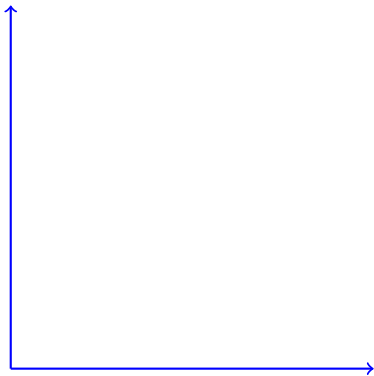
$$x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad x = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \geq 0$$

Example: second basis

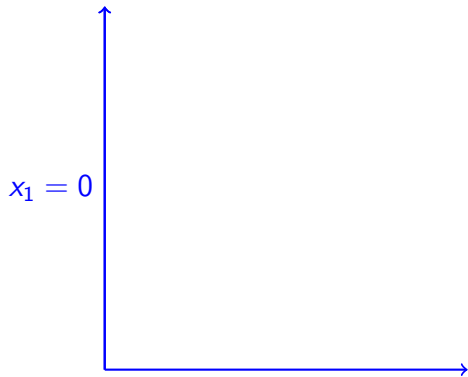
$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad x = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \geq 0$$

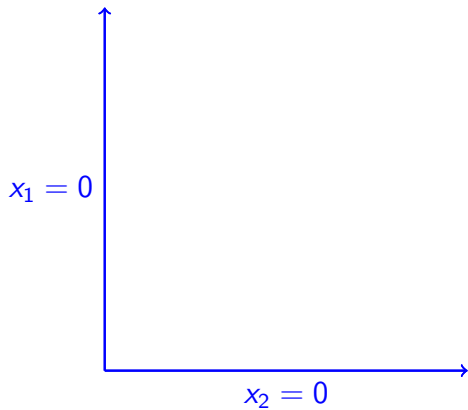
Example



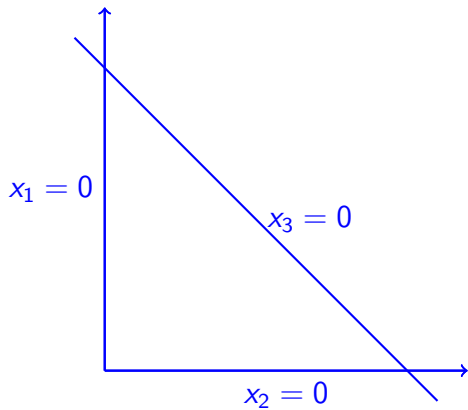
Example



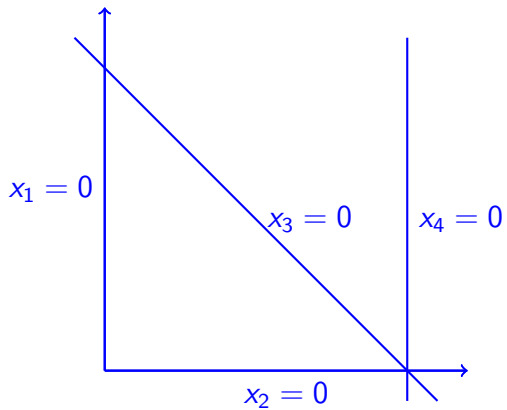
Example



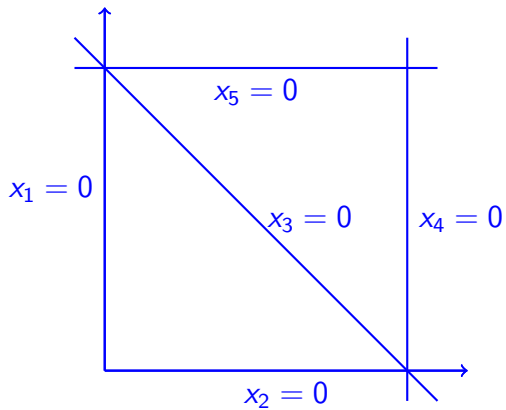
Example



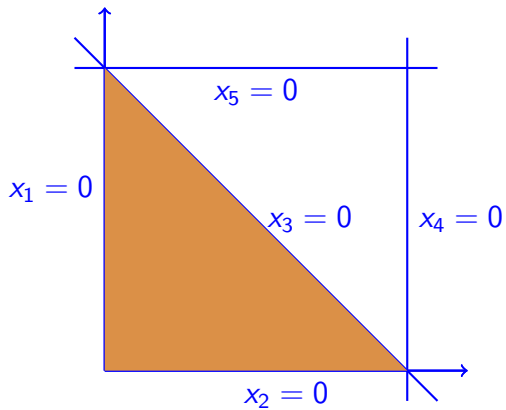
Example



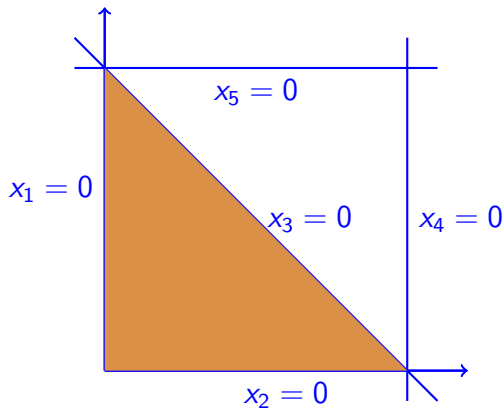
Example



Example



Example



$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Basis 1: [1,3,5]

Basis 2: [1,4,5]

Definition

$$\mathcal{P} = \{x \in \mathbb{R}^n | Ax = b, x \geq 0\}, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m.$$

A basic solution x is degenerate if

- ▶ more than n constraints are active at x , or
- ▶ more than $n - m$ components of x are 0.

Remind that the equality constraints are always active

Summary

*There is not necessarily a bijection between FBS and vertex.
If some basic variables are 0, the same vertex can be obtained from different FBS.
It may be an issue with the algorithm, that will see only the algebraic representation.*